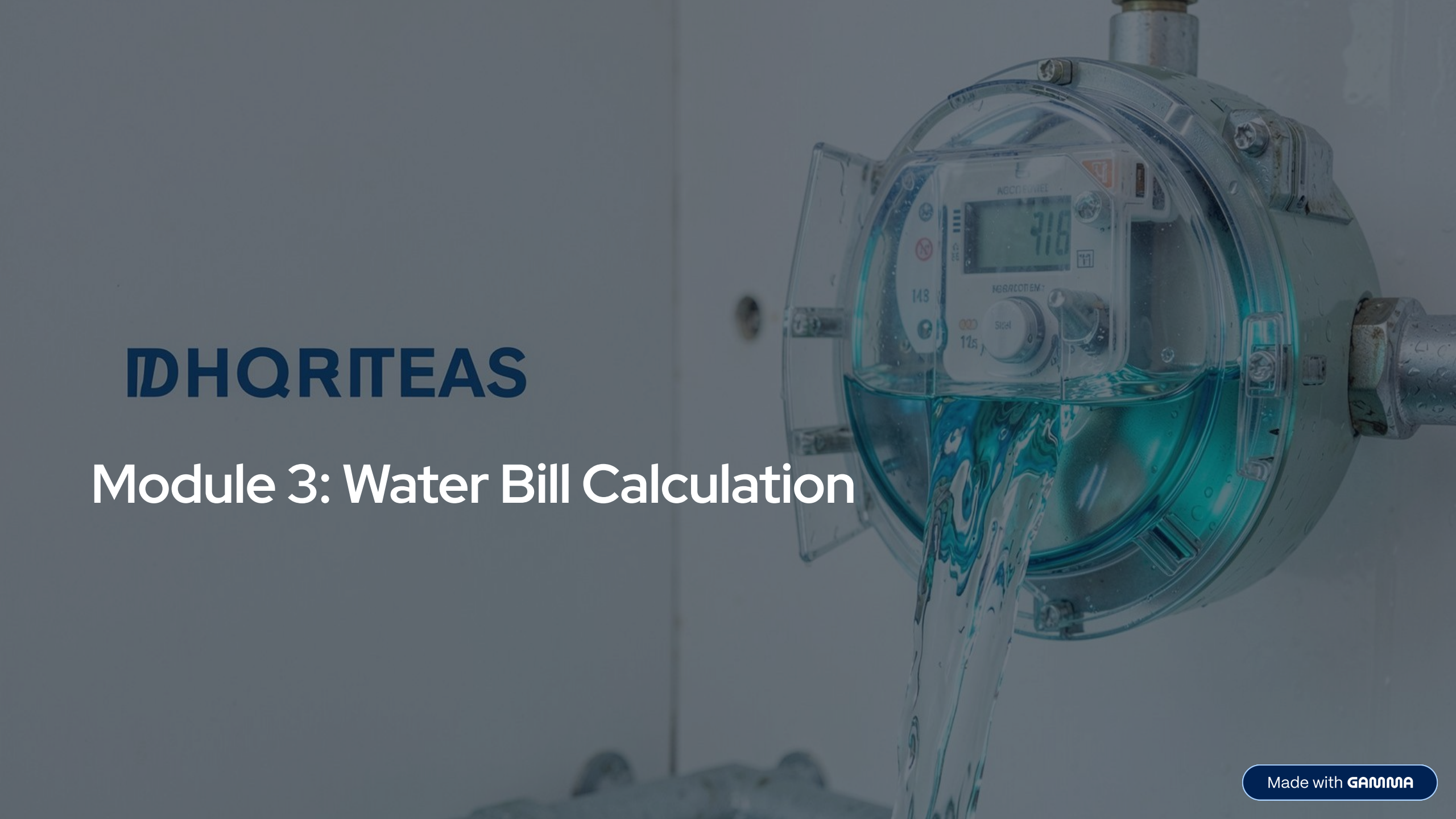


Module 3: Water Bill Calculation

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Objective of Module 3



Calculate the household's water bill.



Apply predefined water-rate slabs.



Use conditional statements for different consumption levels.



Calculate the final payable amount.



Display a clear billing summary.

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Rate Slab System

The application uses different rates depending on the amount of water consumed.

Purpose:

Higher consumption can result in a higher rate, encouraging responsible water usage.

These slab values are project-defined sample rates and can be changed according to the requirements of your application.

WATER CONSUMPTION	RATE
0–100 L	₹2 per L
101–200 L	₹3 per L
201–300 L	₹4 per L
Above 300 L	₹5 per L

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Bill Calculation Logic

The system checks the consumption using if-else statements.

Flow:

01

Water Consumption

02

Check Consumption Range

03

Select Appropriate Rate

04

Calculate Bill

05

Display Final Amount

This makes the billing process automatic.

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Formula

Basic Formula

Water Bill = Water Consumption × Applicable Rate

Example

- Water Consumption = 180 L
- Applicable Rate = ₹3/L
- Bill = 180 × 3
- Total Bill = ₹540

The calculated amount can then be stored as part of the household's billing record.

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Java Implementation

```
double calculateBill(double consumption) {
    double rate;

    if (consumption <= 100) {
        rate = 2;
    } else if (consumption <= 200) {
        rate = 3;
    } else if (consumption <= 300) {
        rate = 4;
    } else {
        rate = 5;
    }

    return consumption * rate;
}
```

Java concepts used:

- Methods
- if-else-if
- Variables
- Relational operators
- Return values

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Billing Example

- Household Usage: 180 L
- Since 180 L falls within the 101–200 L slab:
- Rate = ₹3/L
- Bill = 180 × ₹3
- Final Bill = ₹540

This eliminates manual calculations and reduces billing errors.

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Billing Summary

HOUSEHOLD

ABC Residence

PREVIOUS READING

1250 L

CURRENT READING

1430 L

CONSUMPTION

180 L

APPLICABLE RATE

₹3/L

TOTAL BILL

₹540

This gives the user a clear view of their water usage and bill.

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Advantages

Why automatic billing?



Faster calculation



Consistent rate application



Supports different consumption levels



Reduces human errors



Easy to understand



Can be integrated with billing records

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Module 3 Summary

What did we achieve?

01 Used consumption data from Module 2

02 Created predefined rate slabs

03 Applied conditional statements

04 Calculated the water bill automatically

05 Generated a billing summary

06 Prepared billing information for Module 4

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